

[illegible]

1. Th2 allergic pathway

WHAT YOU SEE

A 'Th2' figure under 'ALLERGIC PATHWAY' releasing IL-4/IL-5/IL-13.

WHAT IT ENCODES

In the ALLERGIC (adaptive) pathway, dendritic cells present allergen to naive T cells, which (under IL-4 and alarmins) differentiate into Th2 cells. Th2 cells secrete the type-2 signature cytokines IL-4, IL-5, and IL-13, orchestrating IgE production, eosinophilia, and mucus/AHR. This phenotype is typically early-onset and atopic.

2. IL-4 queen → IgE class switch

WHAT YOU SEE

A blue 'IL-4' royal with 'class switch' to IgE.

WHAT IT ENCODES

IL-4 drives naive T-cell differentiation toward Th2 and induces B-cell immunoglobulin class-switching to IgE. It signals through the type-I and type-II receptors that share the IL-4Rα chain (the target of dupilumab). IL-4 = the IgE/allergic-commitment cytokine.

BOARD TRAP

Cytokine-to-effect pairing is heavily tested: IL-4 → IgE class switch / Th2 differentiation. Don't confuse it with IL-5 (eosinophils) or IL-13 (mucus/FeNO).

3. IL-5 king → eosinophil survival

WHAT YOU SEE

A red 'IL-5' king over a pile of eosinophils 'eosinophil survival'.

WHAT IT ENCODES

IL-5 is the master cytokine for eosinophils — it drives their maturation in the marrow, recruitment, activation, and survival (anti-apoptosis). Mepolizumab and reslizumab neutralize IL-5; benralizumab targets the IL-5 receptor alpha (IL-5Rα) and additionally depletes eosinophils via ADCC.

BOARD TRAP

IL-5 = eosinophils. Anti-IL-5 biologics only work in patients with ELEVATED blood eosinophils (e.g., ≥300/μL) — they are useless in non-eosinophilic asthma, a classic distractor.

4. IL-13 → mucus + goblet + AHR/FeNO

WHAT YOU SEE

A green 'IL-13' royal with 'mucus + goblet', 'FeNO responsive'.

WHAT IT ENCODES

IL-13 (closely related to IL-4, also using IL-4Rα) drives goblet-cell metaplasia and mucus hypersecretion, airway hyperresponsiveness, subepithelial fibrosis, and — by inducing epithelial iNOS — raises FeNO. It also contributes to IgE switching.

BOARD TRAP

IL-13 = mucus + AHR + FeNO. FeNO tracks IL-13/type-2 activity and predicts response to ICS and to dupilumab (which blocks IL-4Rα and therefore both IL-4 and IL-13).

5. Omalizumab archer (anti-IgE)

WHAT YOU SEE

An archer 'anti-IgE' shooting IgE.

WHAT IT ENCODES

Omalizumab is a humanized monoclonal antibody that binds free IgE at its Cε3 (FcεRI-binding) site, lowering free IgE and down-regulating high-affinity receptors on mast cells/basophils. Indicated for moderate-severe ALLERGIC asthma; dose is based on body weight and baseline total IgE level. Also used in chronic urticaria.

BOARD TRAP

Omalizumab requires documented allergen sensitization AND an IgE level within the dosing range — it is the anti-IgE choice for ALLERGIC asthma and does not fit non-allergic eosinophilic disease.

6. Mast cell sensitization

WHAT YOU SEE

A 'mast cell sensitization' knight near IgE.

WHAT IT ENCODES

IgE produced after class switching binds high-affinity FcεRI receptors on mast cells and basophils ('sensitization'). On re-exposure, allergen cross-links surface IgE, triggering degranulation (histamine, tryptase, leukotrienes) and the early allergic response — the step omalizumab interrupts.

7. Dupilumab blocks both (IL-4Rα)

WHAT YOU SEE

A 'DUPILUMAB' shield 'blocks both' IL-4 and IL-13 via IL-4Rα.

WHAT IT ENCODES

Dupilumab targets the shared IL-4 receptor alpha (IL-4Rα) subunit, thereby blocking BOTH IL-4 and IL-13 signaling at once. It is highly effective in type-2 asthma (especially high FeNO and/or eosinophils), nasal polyposis, atopic dermatitis, and EoE.

BOARD TRAP

Dupilumab can cause a transient rise in blood eosinophils after initiation (usually asymptomatic and self-limited); rarely transient hypereosinophilia/EGPA-like presentations occur — generally benign, but a recognized board point. It is the biologic of choice when both IL-4 and IL-13 effects (high FeNO + eos + comorbid atopy) dominate.

8. CRTh2 / amplifies type 2

WHAT YOU SEE

A 'CRTh2 — amplifies type 2' scroll at the table.

WHAT IT ENCODES

CRTh2 is the prostaglandin D2 (PGD2) receptor on Th2 cells, eosinophils, and ILC2s; PGD2 released from mast cells signals through it to chemoattract and activate these cells, amplifying the type-2 response. CRTh2 antagonists were investigated but have limited clinical role.

9. ILC2 non-allergic pathway

WHAT YOU SEE

An 'ILC2' figure under 'NON-ALLERGIC PATHWAY' with FeNO/AHR.

WHAT IT ENCODES

Group-2 innate lymphoid cells (ILC2) are the INNATE counterpart of Th2 cells: activated directly by the epithelial alarmins TSLP/IL-25/IL-33 (no antigen/IgE needed), they pour out IL-5 and IL-13, producing eosinophilic, high-FeNO asthma that is often late-onset and non-atopic.

BOARD TRAP

Eosinophilic asthma can be NON-allergic (ILC2-driven) — anti-IL-5 biologics still work, but omalizumab (anti-IgE) may NOT, because there is no IgE-mediated allergy to target.

10. Tezepelumab (anti-TSLP) upstream

WHAT YOU SEE

A 'TEZEPELUMAB anti-TSLP' figure blocking the alarmin source.

WHAT IT ENCODES

Tezepelumab blocks TSLP, the most upstream epithelial alarmin, shutting off both the allergic (Th2) and non-allergic (ILC2) branches before they diverge. This gives efficacy across the widest range of phenotypes, including patients with LOW eosinophils and low FeNO.

BOARD TRAP

Tezepelumab is the BROADEST biologic — uniquely effective even with low type-2 biomarkers, making it the answer for severe asthma that is uncontrolled but does not have high eos/IgE/FeNO.

11. Neutrophilic / non-type 2 (steroid-resistant)

WHAT YOU SEE

A dark 'NON-TYPE 2 neutrophilic' cell with IL-17/IL-6/IL-8, 'steroid-resistant'.

WHAT IT ENCODES

Non-type-2 neutrophilic asthma is driven by the Th17 axis (IL-17, IL-6, IL-8/CXCL8) recruiting neutrophils. It is linked to obesity, smoking, and severe/late-onset disease and characteristically responds poorly to inhaled or systemic corticosteroids.

BOARD TRAP

Type-2-targeted biologics (anti-IgE, anti-IL-5, anti-IL-4Rα) do NOT help neutrophilic asthma, and it is steroid-resistant — a frequent exam trap when low eosinophils/FeNO are given.

12. Leukotriene bench (LTRA / 5-LOX)

WHAT YOU SEE

A 'LEUKOTRIENE BENCH' with cys-LTs and 'bronchospasm + mucus'.

WHAT IT ENCODES

Cysteinyl leukotrienes (LTC4/D4/E4), made from arachidonic acid via 5-lipoxygenase, cause potent bronchoconstriction, mucus secretion, vascular leak, and eosinophil recruitment. Montelukast/zafirlukast block the CysLT1 receptor; zileuton inhibits 5-LOX (and needs LFT monitoring).

BOARD TRAP

Leukotriene modifiers are especially useful in AERD (aspirin-exacerbated respiratory disease) and in exercise-induced bronchoconstriction; montelukast carries an FDA neuropsychiatric boxed warning.
